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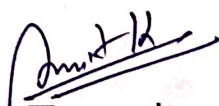
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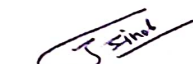
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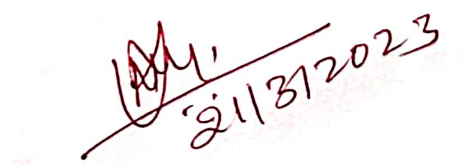
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References:

1. Ajoy Ghatak, "Optics", Publisher Tata Mc Graw Hill, fifth Edition
2. A. Ghatak, K. Thyagarajan, Introduction to Optical Fiber Optics, Publisher, Cambridge University Press.
3. A. Beiser, "Concept of Modern Physics", Publisher, Tata Mc Graw Hill, Fifth Edition.


Faculty


HoD


21/3/2023
Dean Academic

Topic: - Basic Physical Interpretations**Scalar:**

A quantity that is characterized only by magnitude and represented by algebraic sign is called as Scalar.

Eg: Mass, Time, Temperature etc.

Vector:

A quantity that has direction as well as magnitude is called a vector.

Force, velocity, displacement and acceleration are example of vector quantity.

A vector can be geometrically represented by an arrow whose direction is appropriately chosen and whose length is proportional to magnitude of vector.

Field:

If at each point of a region there is a corresponding value of some physical function, the region is called as field.

Field may be classified as either scalar or vector depending on type of function involved.

If the value of physical function at each point is scalar quantity, then the field is scalar field.

The temperature of atmosphere, height of the surface of the earth above sea level are example of scalar field.

When the value of function at each point is a vector quantity the field is called vector quantity.

The wind velocity of the atmosphere,

Force of gravity on mass in space,

And forces on a charged body placed in electric field are examples of vector field.

Gradient, Divergence and Curl:

Gradient of any scalar is the maximum space rate of change of that function.

If the scalar functions V represents temperature

Then $\nabla V = \text{grad } V$ is the temperature gradient or rate of change of temperature with distance.

Temperature gradient ∇V is a vector quantity its direction being that in which temperature changes most rapidly.

The vector quantity may be expressed in terms of its components in the x, y and z direction.

These are respectively

$$\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \text{ and } \frac{\partial V}{\partial z}$$

The resultant temperature gradient is the vector sum of these 3 components

$$\Delta V = \frac{\partial V}{\partial x} \hat{x} + \frac{\partial V}{\partial y} \hat{y} + \frac{\partial V}{\partial z} \hat{z}$$

If suppose V represents electric potential then ∇V represents potential gradient or electric field strength in volts per meter.

Divergence:

The divergence of vector field $\vec{A}$ is represented by $\vec{\nabla} \cdot \vec{A}$. So it is given by

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Clearly divergence of vector field is a scalar. Also, the divergence of a scalar cannot be obtained as the dot product of $\vec{\nabla}$ with a scalar is not possible.

In order to make the clear meaning of divergence, we consider the net flux $\oint \vec{A} \cdot d\vec{s}$ of a vector field $\vec{A}$ from a closed surface S. The divergence of $\vec{A}$ is defined as the net outward flux per unit volume over a closed surface S.

Actually the divergence of $\vec{A}$ at a given point is a measure of how much the vector $\vec{A}$ spreads out that is diverges from the point. Fig 1 shows that divergence of vector field at point O is positive, as the vector spreads out. However in Fig 2, the vector converges and hence the divergence at O is negative.

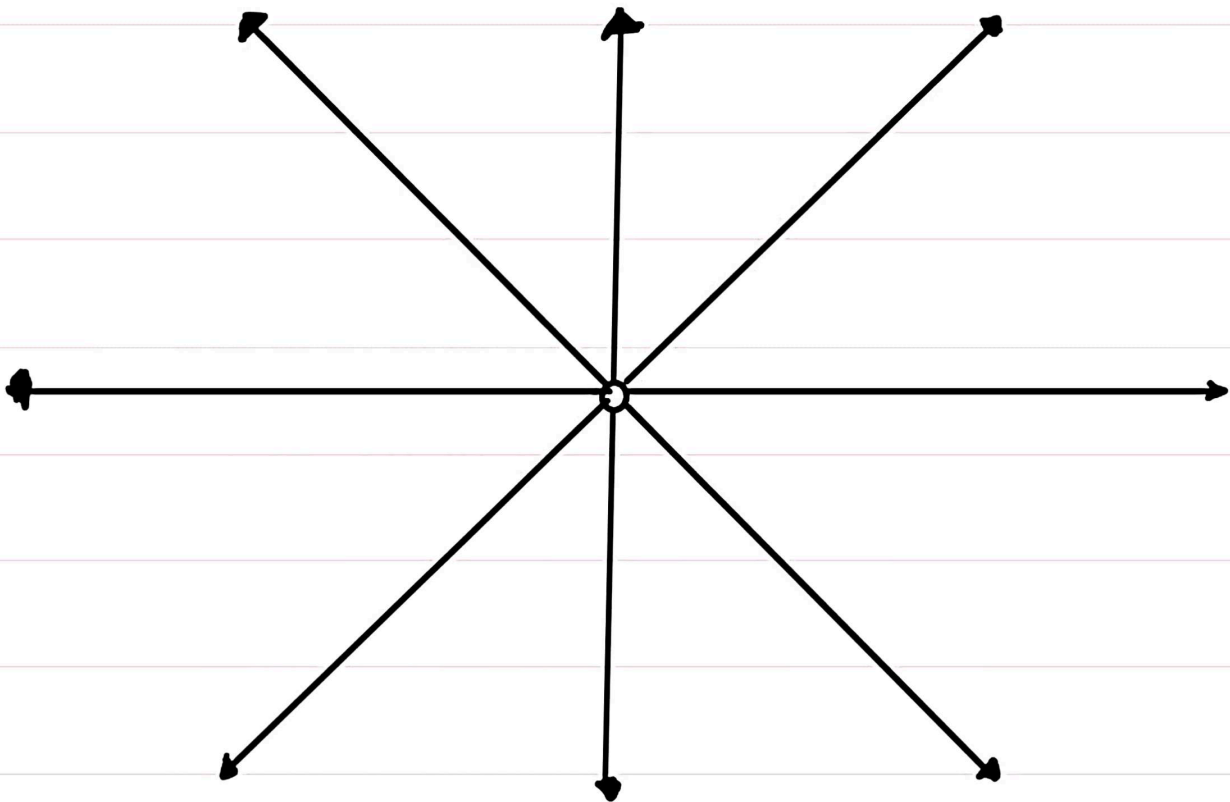


Fig.1

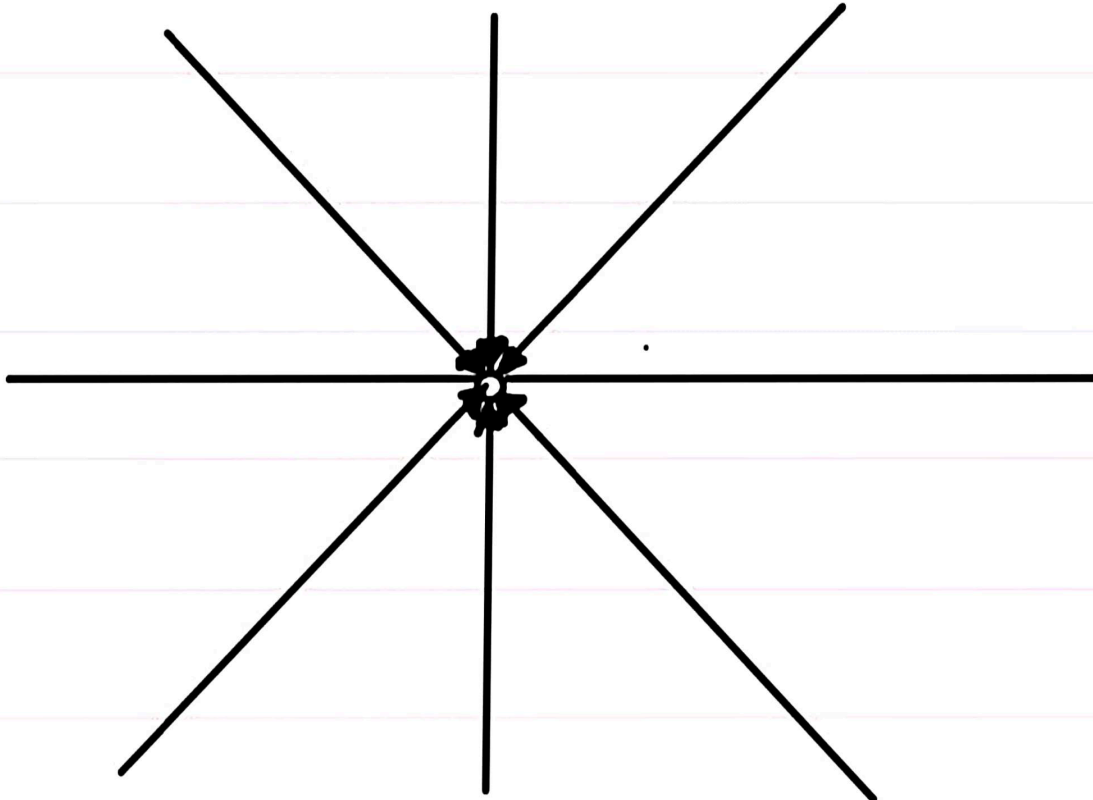


Fig 2 .

Curl:

The curl of vector field $\vec{A}$ is represented by $\vec{\nabla} \times \vec{A}$. So it is given by

$$\vec{\nabla} \times \vec{A} = \hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{j} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

Clearly the curl is a vector quantity, i.e. it has both direction and magnitude. Also it is evident that we cannot have the curl of scalar as the cross product of $\vec{\nabla}$ with a scalar is meaningless.

In order to make clear the meaning of curl, we consider the circulation of vector field $\vec{A}$ around a closed path, i.e. $\oint \vec{A} \cdot d\vec{l}$. It is evident that curl of $\vec{A}$ is a rotational vector.

Vector Identities to Be Used for Derivations:

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$$

Maxwell Equation (Differential Form).	Maxwell's Equation (Integral Form)
① $\nabla \cdot \vec{D} = \rho$	$\rightarrow \oint_S \vec{D} \cdot d\vec{s} = \int_V \rho \cdot dv$
② $\nabla \cdot \vec{B} = 0$	$\rightarrow \oint_S \vec{B} \cdot d\vec{s} = 0$
③ $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\rightarrow \oint_L \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \oint_S \vec{B} \cdot d\vec{s}$
④ $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\rightarrow \oint_L \vec{H} \cdot d\vec{l} = \oint_S (\vec{J} + \frac{\partial \vec{D}}{\partial t}) \cdot d\vec{s}$

$\vec{D}$ represents displacement density $\vec{D} = \epsilon_0 \vec{E}$

$\vec{H} \rightarrow$ Magnetic Field Intensity.

$\vec{B} \rightarrow$ Magnetic Field Density.

$\vec{E} \rightarrow$ Electric Field.

$\rho =$ Volume charge Density.

$\rightarrow$ Application of Maxwell's Equation:

① $\nabla \cdot \vec{D} = \rho$

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V \rho \, dv$$

$$\oint_S \epsilon_0 \vec{E} \cdot d\vec{s} = \int_V \rho \, dv$$

$$\oint \vec{E} \cdot d\vec{s} = \int \rho \, dv$$

$$\oint_S \vec{E} \cdot d\vec{s} = \int_V \frac{\rho}{\epsilon_0} dv$$

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

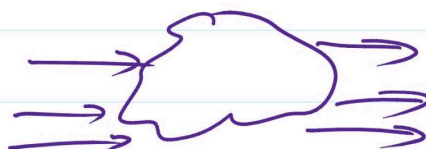
So it is clearly visible that the above equation represents Gauss law of Electrostatics.

②

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\oint_S \vec{B} \cdot d\vec{s} = 0$$

$$(\vec{B} \cdot \vec{A}) = \phi$$



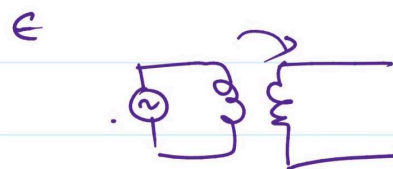
clearly this equation states that magnetic flux across the closed surface is always zero, Hence second equation represents Gauss law of Magnetostatics.

③

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\epsilon = -\frac{d\phi}{dt}$$

→ When we carefully look at this equation it is stating that the emf induced is nothing but rate of change of magnetic flux. that

$$e = -\frac{d\phi}{dt}$$


where minus sign represents opposition.

④

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Modified form of Ampere Circuital law.

$$\oint \vec{H} \cdot d\vec{\ell} = \int (\vec{J} + \frac{\partial \vec{D}}{\partial t}) \cdot d\vec{S}$$

$$\oint \mu_0 \vec{H} \cdot d\vec{\ell} = \mu_0 \int \vec{J} \cdot d\vec{S}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \int \vec{J} \cdot d\vec{S}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I \quad \text{Ampere Circuital law}$$

Eq ④ of Maxwell represents Modified Form of Ampere Circuital law.

Hence we can summarize,

$$\vec{\nabla} \cdot \vec{D} = \rho : \text{Gauss law of Electrostatics.}$$

$$\vec{\nabla} \cdot \vec{B} = 0 : \text{Gauss law of Magnetostatics.}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} : \text{Faraday's law of Electromagnetic Induction.}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} : \text{Modified form of Ampere Circuital law.}$$

$$\textcircled{1} \quad \vec{\nabla} \cdot \vec{D} = \rho$$

$$\textcircled{2} \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\textcircled{3} \quad \vec{\nabla} \times \vec{E} = -\frac{\delta \vec{B}}{\delta t}$$

$$\textcircled{4} \quad \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\textcircled{1} \quad \vec{\nabla} \cdot \vec{D} = \rho$$

As we know from Gauss law of Electrostatics, electric flux through a surface S is equal to charge enclosed in the surface.

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} \quad - \textcircled{1}$$

Now re arranging the above equation over a volume dv the equation becomes

$$\oint_S (\underbrace{\epsilon_0 \vec{E}}_{\vec{D}}) \cdot d\vec{s} = \int_V \rho \cdot dv \quad - \textcircled{2}$$

Now Since we know that $\epsilon_0 \vec{E}$ represents Displacement density $\vec{D}$, hence the above equation becomes.

$$\textcircled{1} \oint_S \vec{D} \cdot d\vec{s} = \int_V \rho \cdot dv \quad - \textcircled{3}$$

Now it can be clearly seen that on one side there is surface integral and on one side volume integral, hence to evaluate, we need to convert surface integral into volume integral. Applying divergence theorem.

$$\int_V (\vec{\nabla} \cdot \vec{D}) \cdot dv = \int_V \rho \cdot dv$$

$$\int_V (\vec{\nabla} \cdot \vec{D} - \rho) \cdot dv = 0$$

Now differentiating both sides we get

Now differentiating both sides we get

$$\vec{\nabla} \cdot \vec{D} - \rho = 0$$

$$\boxed{\vec{\nabla} \cdot \vec{D} = \rho} \rightarrow \text{Proved.}$$

$$\textcircled{2} \quad \vec{\nabla} \cdot \vec{B} = 0$$

As we know from Gauss law of Magnetostatics, that magnetic field over a closed surface is always zero hence we can write mathematically

as

$$\Phi_B = \oint \vec{B} \cdot d\vec{s} = 0 \quad - \textcircled{1}$$

Now applying Gauss Divergence Theorem, surface integral can be converted into volume integral so the eq now becomes.

$$\int (\vec{\nabla} \cdot \vec{B}) \cdot dV = 0$$

Differentiating both sides

$$\boxed{\vec{\nabla} \cdot \vec{B} = 0} \quad - \textcircled{2}$$

$$\textcircled{3} \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

From Faraday's law of Electromagnetic Induction EMF induced is defined as rate of change of flux and is given by following mathematical equation.

$$\text{EMF} = -\frac{\partial \Phi}{\partial t} \quad - \textcircled{1}$$

Also we know that EMF and electric field over line element $d\vec{e}$ is related as

$$\text{EMF} = \oint \vec{E} \cdot d\vec{e} \quad - \textcircled{2}$$

$$\text{EMF} = \oint \vec{E} \cdot d\vec{\ell} \quad - (1)$$

Since eq (1) and eq (2) both represent emf we can write,

$$\oint \vec{E} \cdot d\vec{\ell} = - \frac{\partial \Phi}{\partial t}$$

$$\Phi = BA$$

$$\oint \vec{B} \cdot d\vec{s}$$

$$\oint \vec{E} \cdot d\vec{\ell} = - \frac{\partial}{\partial t} \left(\oint \vec{B} \cdot d\vec{s} \right)$$

Now it can be clearly seen that one side of the equation we have line integral and other side we have surface integral. so we need to convert line integral into surface integral for this use Stokes curl theorem.

$$\oint (\nabla \times \vec{E}) \cdot d\vec{s} = - \frac{\partial}{\partial t} \oint \vec{B} \cdot d\vec{s}$$

$$\boxed{\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}}$$

$$(4) \quad (\nabla \times \vec{H}) = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

As we know from ampere circuital law that the line integral of magnetic field density over an element $d\ell$ is μ_0 times the current enclosed. This could be mathematically represented as.

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I \quad - (1)$$

$$\oint (\cancel{\mu_0} \vec{H}) \cdot d\vec{\ell} = \cancel{\mu_0} I$$

$$\oint \vec{H} \cdot d\vec{\ell} = I \quad - (2)$$

$$J = \frac{I}{A}$$

eq (2) can now be written as

$$\oint \vec{H} \cdot d\vec{\ell} = \oint \vec{J} \cdot d\vec{s}$$

$$I = J(A)$$

$$\oint \vec{J} \cdot d\vec{s}$$

$$\oint \vec{H} \cdot d\vec{\ell} = \oint \vec{J} \cdot d\vec{s}$$

$$\oint \vec{J} \cdot d\vec{s}$$

Apply Stoke's Curl theorem on LHS we get

$$\oint (\vec{\nabla} \times \vec{H}) \cdot d\vec{s} = \oint \vec{J} \cdot d\vec{s} \quad - (3)$$

Applying differentiation on both sides.

$$\vec{\nabla} \times \vec{H} = \vec{J}$$

$$- (4) \quad \text{Diagram of a cube with a vector field passing through it.}$$

To check the above equation for Maxwell's continuity eq. take the div of eq (4).

$$i = -\frac{dq}{dt}$$

$$J \cdot A = -\frac{dq}{dt}$$

$$\oint \vec{J} \cdot d\vec{s} = -\frac{d}{dt} \int \rho dv$$

$$\boxed{\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}}$$

Maxwell's Continuity eq.



$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} \quad - (5)$$

$$\boxed{\vec{\nabla} \cdot \vec{J} = 0}$$

→ (6)

Since eq (6) doesn't follow Maxwell's Continuity equation, then Maxwell did some modification and the law is termed as

Modified form of ampere circuital law.

and Modified eq (4)

$$\vec{\nabla} \times \vec{H} = \vec{J} + \vec{J}_d \rightarrow \text{Displacement current density.} \quad - (7)$$

Applying div in eq (7) in both sides.



$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H}) = \vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \vec{J}_d$$

$$\vec{\nabla} \cdot \vec{J} = -\vec{\nabla} \cdot \vec{J}_d$$

Now from Maxwell's Continuity Equation.

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\vec{\nabla} \cdot \vec{J} = -\vec{\nabla} \cdot \vec{J}_d$$

$$\cancel{\frac{\partial \rho}{\partial t}} = \cancel{\vec{\nabla} \cdot \vec{J}_d}$$

$$\vec{\nabla} \cdot \vec{J}_d = \frac{\partial \rho}{\partial t} \quad - (8)$$

Now from first equation of Maxwell i.e.

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Hence substituting in eq (8) we get,

$$\vec{\nabla} \cdot \vec{J}_d = \frac{\partial (\vec{\nabla} \cdot \vec{D})}{\partial t} \quad - (9)$$

eq (9) can now be modified as.

$$\vec{J}_d = \frac{\delta \vec{D}}{\delta t} \quad - (10)$$

Substituting eq (10) in eq (7) we have,

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\delta \vec{D}}{\delta t}$$

Proved
4th Eq. of Maxwell is proved

Derivation of Maxwell's Equation in Integral Forms

$$\textcircled{1} \quad \vec{\nabla} \cdot \vec{D} = \rho \quad - \textcircled{1}$$

On integrating equation (1) over a volume dv we have.

$$\int_V (\vec{\nabla} \cdot \vec{D}) \cdot dv = \int_V \rho \cdot dv$$

Using Gauss Divergence Theorem on LHS, we get

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V \rho \cdot dv$$

(or)

$$\oint_S \vec{D} \cdot d\vec{s} = q$$

$$\nabla \times (\nabla \times F) = \nabla(\nabla \cdot F) - \nabla^2 F$$

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H})$$

Maxwell's Equation (Differential Form):

$$1. \vec{\nabla} \cdot \vec{D} = \rho$$

$$2. \vec{\nabla} \cdot \vec{B} = 0$$

$$3. \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$4. \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Maxwell's Equation (Integral Form):

$$1. \oint_S \vec{D} \cdot d\vec{s} = \int_V \rho \cdot dv$$

$$2. \oint_S \vec{B} \cdot d\vec{s} = 0$$

$$3. \oint_l \vec{B} \cdot d\vec{l} = -\frac{d}{dt} \oint_S \vec{B} \cdot d\vec{s}$$

$$4. \oint_l \vec{H} \cdot d\vec{l} = \oint_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$$

Here D represents displacement density, ρ represents volume charge density, E represents electric field intensity, J represents current density, H represents Magnetic field intensity, and $\frac{\partial \vec{D}}{\partial t}$ represents displacement current.

Applications of Maxwell's Equation:

Consider the Maxwell first equation

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Where $\vec{D}$ represents displacement density and ρ represents volume charge density.

Also we know that

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\textcircled{4} \quad \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad - \textcircled{4}$$

On integrating over a surface ds bounded by closed path we have

$$\oint_s (\vec{\nabla} \times \vec{H}) \cdot d\vec{s} = \oint_s \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}.$$

Applying Stokes's Curl Theorem on LHS we get,

$$\oint_{\text{L}} \vec{H} \cdot d\vec{\ell} = \oint_s \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$$

↳ Integral Form of Maxwell
4th Equation.

→ Maxwell's Equation in Free Space:

$$\textcircled{1} \quad \vec{\nabla} \cdot \vec{D} = \rho$$

$$\textcircled{2} \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\textcircled{3} \quad \vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\textcircled{4} \quad \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Particular case of electromagnetic phenomenon in free space (or) more generally in perfect dielectric containing no charges and no conduction current.

→ So from this we can say that volume charge density ρ and conduction current density will become zero, hence equation of Maxwell in modified form are.

①	$\vec{\nabla} \cdot \vec{D} = 0$
②	$\vec{\nabla} \cdot \vec{B} = 0$
③	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
④	$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$

→ Maxwell's Equation in free space.

Electromagnetic Wave Equation in Free Space in Terms of E.

The equations in free space are.

$$\left\{ \begin{array}{ll} \vec{\nabla} \cdot \vec{D} = 0 & (\vec{\nabla} \cdot \epsilon_0 \vec{E} = 0 \text{ (or) } \vec{\nabla} \cdot \vec{E} = 0) \\ \vec{\nabla} \cdot \vec{B} = 0 & (\vec{\nabla} \cdot \mu_0 \vec{H} = 0 \text{ (or) } \vec{\nabla} \cdot \vec{H} = 0) \\ \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \end{array} \right.$$

The wave Equation is given as

$$\psi \rightarrow \text{psi}$$

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

Wave Equation in terms of E.

$$\underline{\underline{\nabla^2 \vec{E}}} = \frac{1}{v^2} \frac{\partial^2 \vec{E}}{\partial t^2}$$

Now from eq ③ of Maxwell we have

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad - (1)$$

Taking curl on both the sides of Equation we get,

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = - \frac{\partial (\vec{\nabla} \times \vec{B})}{\partial t}$$

$$\underbrace{\vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E}}_0 = - \frac{\partial (\vec{\nabla} \times \vec{B})}{\partial t} \quad - (2)$$

$$- \vec{\nabla}^2 \vec{E} = - \frac{\partial (\vec{\nabla} \times \vec{B})}{\partial t}$$

$$[\because B = \mu_0 H]$$

$$- \vec{\nabla}^2 \vec{E} = - \mu_0 \frac{\partial (\vec{\nabla} \times \vec{H})}{\partial t}$$

$$\left[\because \text{from 4th eq of Maxwell} \right. \\ \left. \vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \right]$$

$$-\nabla^2 \vec{E} = -\mu_0 \frac{\partial}{\partial t} \left(\frac{\partial \vec{D}}{\partial t} \right)$$

$$\therefore \vec{D} = \epsilon_0 \vec{E}$$

$$\cancel{-\nabla^2 \vec{E}} = \cancel{-\mu_0 \epsilon_0} \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\boxed{\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}}$$

→ Maxwell's wave Equation in terms of E.

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

$$\psi = E, \quad \frac{1}{\cancel{v^2}} = \mu_0 \epsilon_0$$

↓ velocity

$$v^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \times \frac{\sqrt{4\pi}}{\sqrt{4\pi}}$$

$$v = \frac{\sqrt{4\pi}}{\sqrt{4\pi \mu_0 \epsilon_0}}$$

$$v = \frac{1}{\sqrt{4\pi \epsilon_0}} \times \frac{\sqrt{4\pi}}{\sqrt{\mu_0}}$$

$$\begin{aligned} \frac{1}{4\pi \epsilon_0} &= 9 \times 10^9 &= (9 \times 10^9)^{1/2} \times \frac{\sqrt{4\pi}}{\sqrt{4\pi \times 10^{-7}}} \\ \frac{1}{\sqrt{4\pi \epsilon_0}} &= (9 \times 10^9)^{1/2} &= (9 \times 10^9)^{1/2} \times (10^7)^{1/2} \\ & &= (9 \times 10^{16})^{1/2} \end{aligned}$$

$$\boxed{v = 3 \times 10^8 \text{ m/s}} \checkmark$$

Electromagnetic Wave Equation in Free Space in terms of H.

Equation in free space.

$$\textcircled{1} \quad \vec{\nabla} \cdot \vec{D} = 0$$

$$\textcircled{2} \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\textcircled{3} \quad \vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\textcircled{4} \quad \vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

From eq $\textcircled{4}$ of Maxwell in free space.

$$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \quad - \textcircled{1}$$

Taking curl on Both sides of the equation

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{H}) = \frac{\partial (\vec{\nabla} \times \vec{D})}{\partial t}$$

using vector
Identity.

$$\underbrace{\vec{\nabla} \cdot (\vec{\nabla} \times \vec{H})}_0 - \vec{\nabla}^2 \vec{H} = \frac{\partial (\vec{\nabla} \times \vec{D})}{\partial t}$$

$$- \vec{\nabla}^2 \vec{H} = \frac{\partial (\vec{\nabla} \times \vec{D})}{\partial t}$$

$$\therefore \vec{D} = \epsilon_0 \vec{E}$$

$$- \vec{\nabla}^2 \vec{H} = \frac{\partial (\vec{\nabla} \times \epsilon_0 \vec{E})}{\partial t}$$

$$- \vec{\nabla}^2 \vec{H} = \epsilon_0 \frac{\partial (\vec{\nabla} \times \vec{E})}{\partial t}$$

Now from eq ③ of maxwell, we have.

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$- \vec{\nabla}^2 \vec{H} = - \epsilon_0 \frac{\partial}{\partial t} \left(\frac{\partial \vec{B}}{\partial t} \right)$$

$$\therefore \vec{B} = \mu_0 \vec{H}$$

$$\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2}$$

$$\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2}$$

Comparing these
equation.

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

$$\psi = \vec{H}, \quad \frac{1}{v^2} = \mu_0 \epsilon_0$$

→ Poynting's Theorem:

→ In a uniform plane wave at any point product of Electric Field $\vec{E}$ and magnetic Field Intensity ($\vec{H}$) is a measure of energy flow at that point.

$$\vec{P} = \vec{E} \cdot \vec{H}$$

Electromagnetism

Derivation of Maxwell Equations (Integral Form):

As we know from differential form of first equation that

$$\vec{\nabla} \cdot \vec{D} = \rho \quad (1)$$

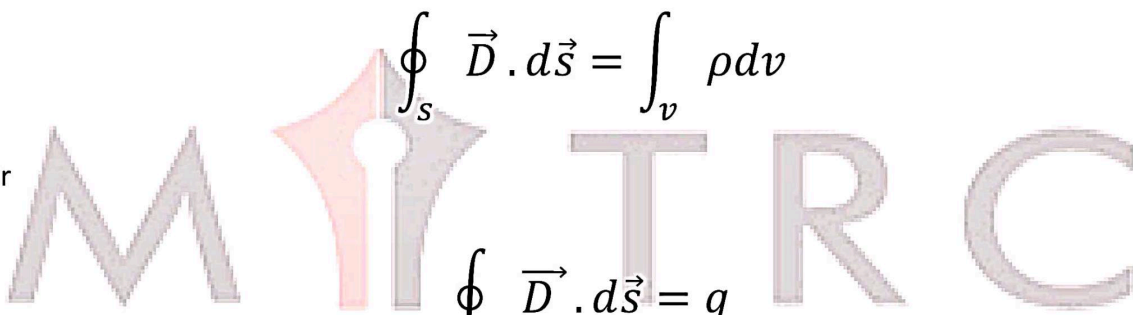
on integrating equation (1) over a volume dv we have

$$\int_v (\vec{\nabla} \cdot \vec{D}) dv = \int_v \rho dv$$

Using Gauss Divergence theorem we have

Or

$$\oint_s \vec{D} \cdot d\vec{s} = \int_v \rho dv$$

$$\oint_s \vec{D} \cdot d\vec{s} = q$$


Here q is the net charge contained in volume V and S is the surface bounding the volume V . hence first equation of Maxwell in integral form is proved.

Maxwell Second Equation:

From Maxwell second equation in differential form we have,

$$\vec{\nabla} \cdot \vec{B} = 0$$

On integrating above equation over a volume dv we have

$$\int_v (\vec{\nabla} \cdot \vec{B}) \cdot dv = 0$$

Using Gauss Divergence theorem we have

$$\oint_s \vec{B} \cdot d\vec{s} = 0$$

Which is the integral form of second equation of Maxwell,

Maxwell Third Equation:

From Maxwell third equation in differential form we have

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

On integrating over a surface ds bounded by closed path we have

$$\oint_s (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} = -\oint_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

Converting surface integral into line integral by Stoke's curl theorem

$$\oint_l \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \oint_s \vec{B} \cdot d\vec{s}$$

Which is integral form of Maxwell's third equation

Maxwell Fourth Equation:

Differential form of Maxwell equation is given is,

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

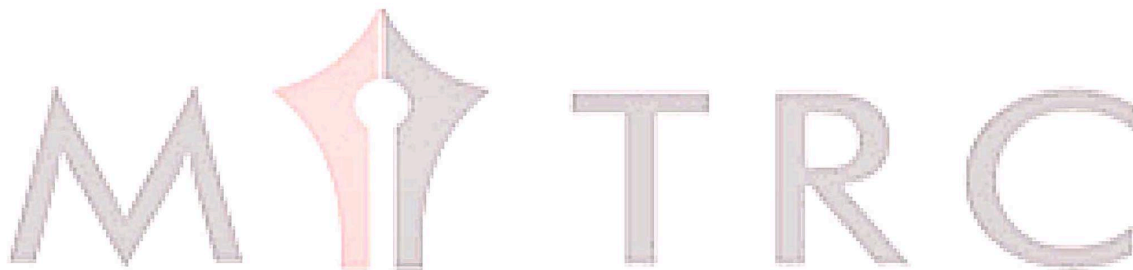
On integrating over a surface ds bounded by a closed path we have,

$$\oint_s (\vec{\nabla} \times \vec{H}) \cdot d\vec{s} = \oint_s \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$$

Converting surface integral into line integral,

$$\oint_l \vec{H} \cdot d\vec{l} = \oint_s \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$$

Which is the integral form of fourth equation of Maxwell.



We know that Maxwell Equation is given by:

$$1. \vec{\nabla} \cdot \vec{D} = \rho$$

$$2. \vec{\nabla} \cdot \vec{B} = 0$$

$$3. \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$4. \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Particular case of electromagnetic phenomenon in free space or more generally in perfect dielectric **containing no charges and no conduction current.**

So from this we can say that volume charge density ρ and conduction current density will become zero hence the equations of Maxwell in modified form are:

$$\vec{\nabla} \cdot \vec{D} = 0$$

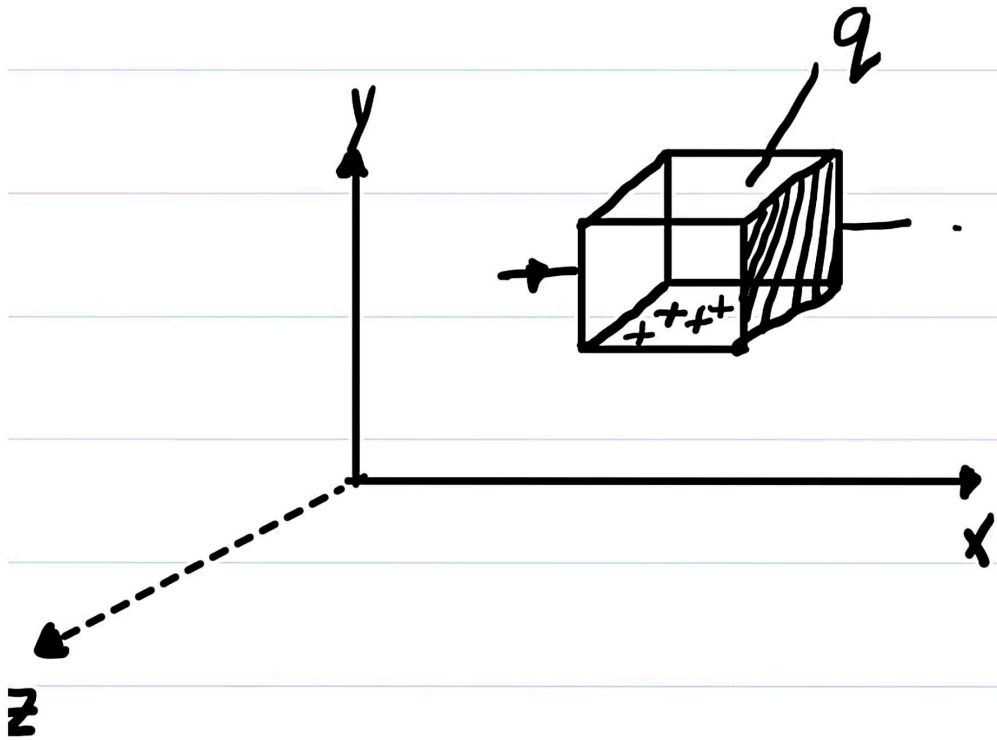
$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

So the above equations are Maxwell equations in free space.

Maxwell's Continuity Equation and Concept of Displacement Current:



It is based on conservation of charge.

Current is simply charge in motion.

Total current flowing out of the volume must be equal to rate of decrease of charge within the molecule.

Let q charges in a cube of volume V come out of the surface. If there is flow of charge that means current will be there.

As current flows there will be decrease in charges present inside the cube w.r.t to time

$$I = -\frac{dq}{dt}$$

When we consider charge over a volume v

$$q = \int_v \rho dv$$

And considering current over surface ds we have,

$$I = \oint_s \vec{j} \cdot d\vec{s}$$

From law of conservation of charge

$$\oint_s \vec{J} \cdot d\vec{s} = -\frac{d}{dt} \int_v \rho dv$$

Applying Gauss Divergence Theorem we have,

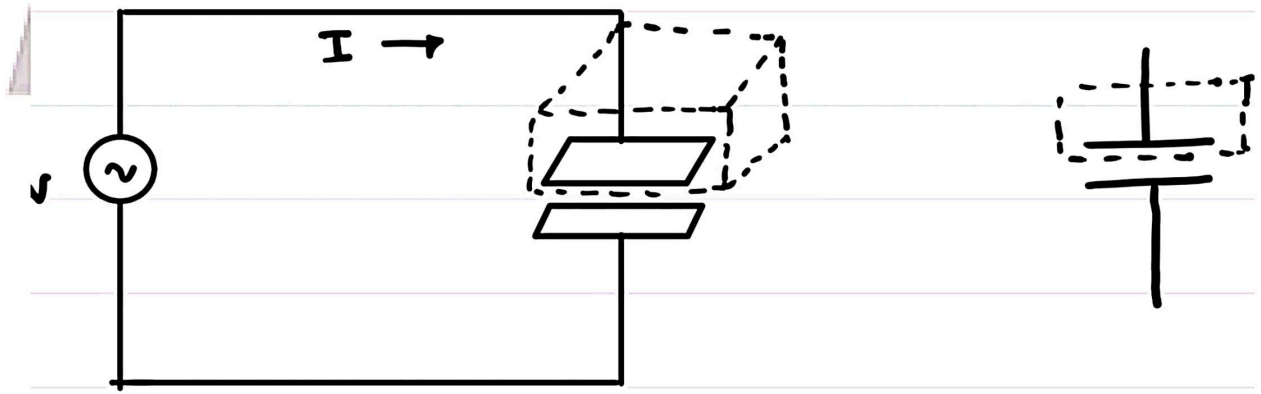
$$\int_v (\vec{\nabla} \cdot \vec{J}) dv = -\frac{d}{dt} \int_v \rho dv$$

$$\int_v \left[\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} \right] dv = 0$$

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

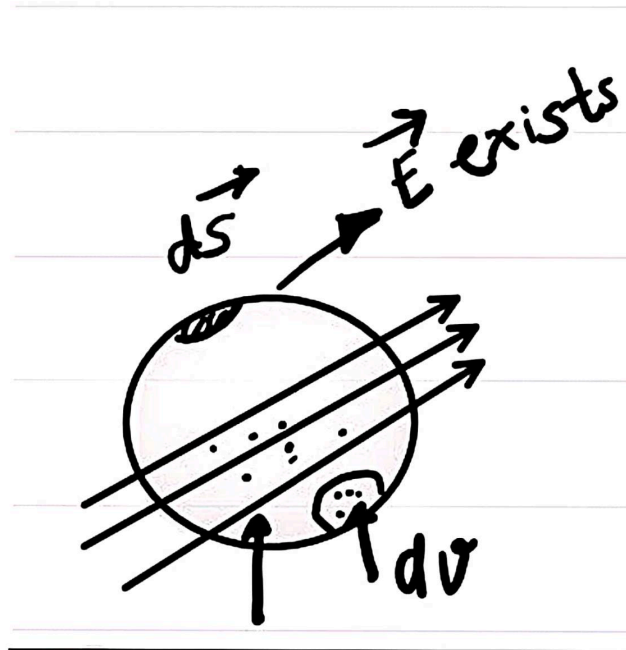
This is the equation of continuity.

Displacement Current:



Maxwell assumed that a changing displacement density (that is changing Electrical Field) was equivalent to electric current density and as such would produce a magnetic field, has had most far reaching effects.

The current which is produced due to changing displacement density is termed as displacement current. And the current flowing in the conductor is called as conduction current.

Divergence and Curl of Electrostatic Field:

As we know from Gauss Law of Electrostatics, electric flux through a closed surface S is equal to charge enclosed in the surface, which could be mathematically represented as

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

Now rearranging above equation and considering the equation over a volume dv the equation becomes,

$$\oint_S (\epsilon_0 \vec{E}) \cdot d\vec{S} = \int_V \rho \cdot dv$$

It is clearly evident that on LHS we have surface integral and on RHS we have volume integral, hence we need to convert surface integral into volume integral using **Gauss Divergence Theorem** we have,

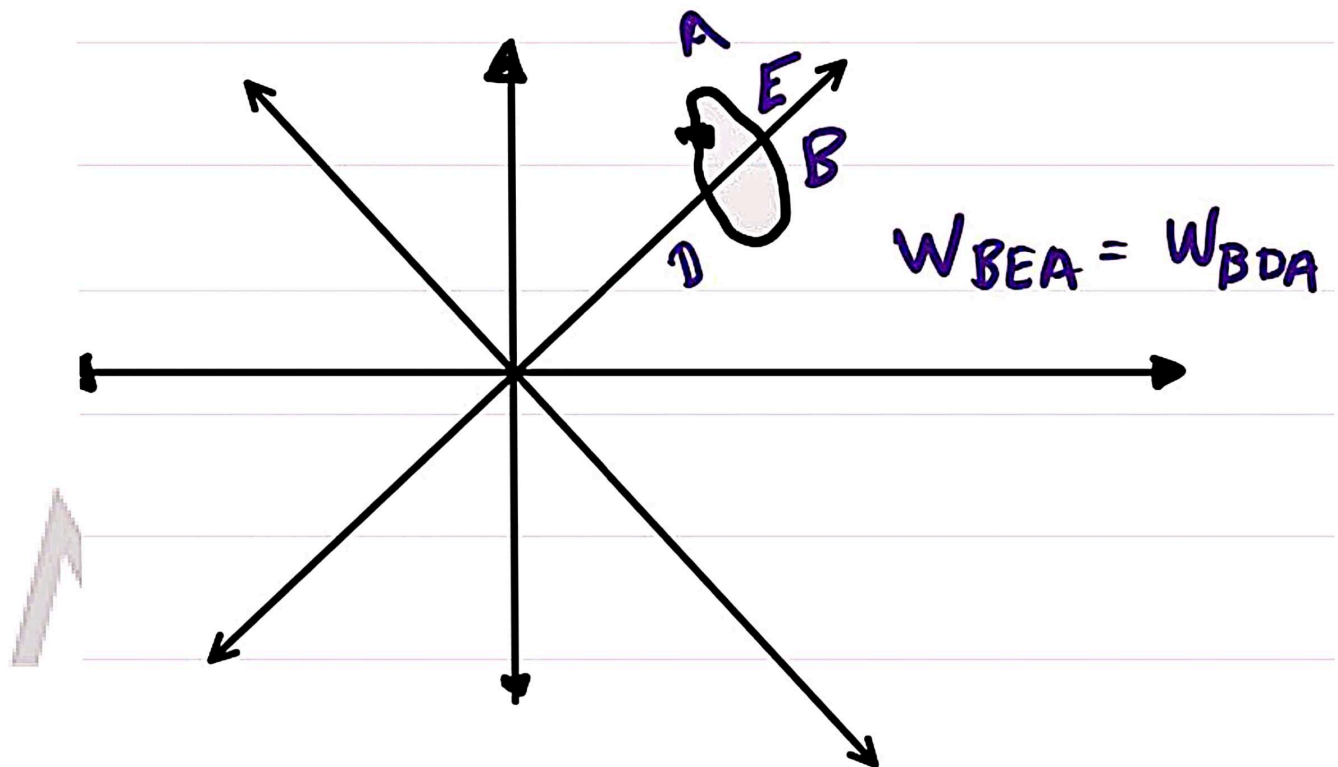
$$\int_V (\vec{\nabla} \cdot \vec{E}) \cdot dv = \int_V \frac{\rho}{\epsilon_0} \cdot dv$$

Hence we have,

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Which means that divergence of electrostatic field is volume charge density per unit epsilon.

Curl of Electrostatic Field:



As we know that electrostatic field is a conservative force. The work done in a conservative force will be zero. We know that work done is force over a particular distance, when we consider small element $d\vec{l}$, the total work done is given by following expression,

$$\oint_l (\vec{F} \cdot d\vec{l}) = 0 \quad (1)$$

The force due to a static charge is given by

$$\vec{F} = q\vec{E}$$

Substituting in equation 1 we have

$$\oint_l \vec{E} \cdot d\vec{l} = 0$$

Applying Stoke's Curl theorem line integral can be converted into surface integral we have,

$$\oint_s (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} = 0$$

Hence the above equation can be rewritten as,

$$\vec{\nabla} \times \vec{E} = 0$$

Hence from the above equation we can write that the curl of electrostatic field is zero over a closed surface.

Divergence and Curl of Magnetic Field:

As we know from **Gauss law of Magnetostatics**, that the magnetic flux enclosed over a closed surface is always zero hence we can write mathematically as

$$\phi_B = \oint_s \vec{B} \cdot d\vec{s} = 0$$

Now applying **Gauss Divergence Theorem**, surface integral can be converted into volume integral so the equation now becomes,

$$\int_v (\vec{\nabla} \cdot \vec{B}) dv = 0$$

Therefore from mathematics we have

$$\vec{\nabla} \cdot \vec{B} = 0$$

Hence we can say that divergence of magnetic field is zero over a closed surface.

Curl of Magnetic Field:

As we know from ampere's circuital law that the line integral of magnetic field density over an element dl is μ_o times the current enclosed. This could be mathematically represented as,

$$\oint_l \vec{B} \cdot d\vec{l} = \mu_o I$$

Also we know that $\vec{B} = \mu_o \vec{H}$ also considering current density over a closed surface we have

$$\oint_l \mu_o \vec{H} \cdot d\vec{l} = \mu_o \oint_s \vec{J} \cdot d\vec{s}$$

Using Stoke's Curl theorem line integral into surface integral can be given as follows,

$$\oint_s (\vec{\nabla} \times \vec{B}) \cdot d\vec{s} = \oint_s \mu_o \vec{J} \cdot d\vec{s}$$

From the above equation we can conclude that

$$\vec{\nabla} \times \vec{B} = \mu_o \vec{J}$$

The above equation states that curl of magnetic field is μ_o times the current density.

Poisson's and Laplace Equations:

As we know that Electric field is the region around a charge in which another charge experience force.

The electrostatic field $\vec{E}$ is a special kind of field whose curl is always zero.

Since any vector whose curl is zero is equal to gradient of some scalar quantity as electric potential V.

$$\int (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} = \oint \vec{E} \cdot d\vec{l} = 0$$

It means that the line integral of $\vec{E}$ from point a1 to point a2 will be same for all paths between these points. Line integral is independent of the path.

$$V = - \int_a^r \vec{E} \cdot d\vec{l}$$

$$\vec{E} = -\vec{\nabla}V$$

V is called as electric potential.

Gauss Law in Linear medium is written as

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Also we have seen that

$$\vec{E} = -\vec{\nabla}V$$

Also we know that $\vec{B} = \mu_o \vec{H}$ also considering current density over a closed surface we have

$$\oint_l \mu_o \vec{H} \cdot d\vec{l} = \mu_o \oint_s \vec{J} \cdot d\vec{s}$$

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$$V = - \int_a^r \vec{E} \cdot d\vec{l}$$

$$\vec{E} = -\vec{\nabla}V$$

V is called as electric potential.

Gauss Law in Linear medium is written as

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Also we have seen that

$$\vec{E} = -\vec{\nabla}V$$

Therefore we have

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}$$

The above equation is called as Poisson's Equation.

For a free charge region we have $\rho = 0$

Then Poisson equation takes the form,

$$\nabla^2 V = 0$$

This is known as Laplace Equation.

Faraday's Law of Electromagnetic Induction: Faraday's law of electromagnetic induction, also known as Faraday's law is the basic law of electromagnetism which helps us to predict how a magnetic field would interact with an electric circuit to produce an electromotive force (EMF). This phenomenon is known as electromagnetic induction.

The law was proposed in the year 1831 by an experimental physicist and chemist named Michael Faraday. So you can see where the name of the law comes from. That being said, the Faraday's law or the law of electromagnetic induction is basically the results or the observations of the experiments that Faraday conducted. He performed three main experiments to discover the phenomenon of electromagnetic induction

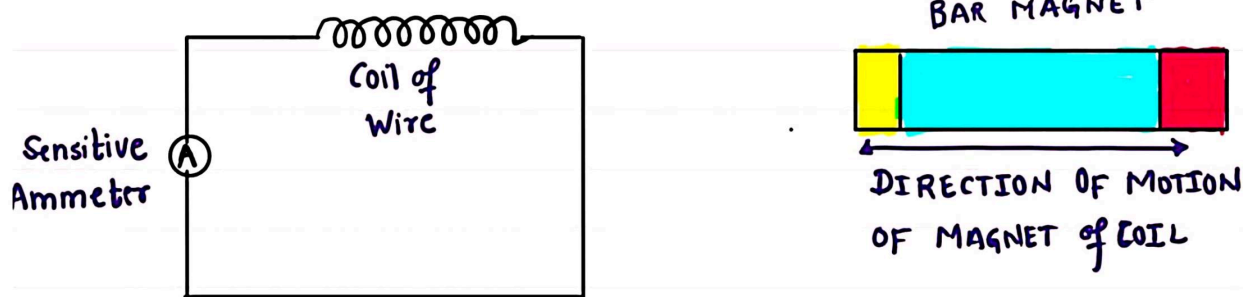
Faraday's First Law of Electromagnetic Induction

The discovery and understanding of electromagnetic induction are based on a long series of experiments carried out by Faraday and Henry. From the experimental observations, Faraday arrived at a conclusion that an emf is induced in the coil when the magnetic flux across the coil changes with time. With this in mind, Faraday formulated his first law of electromagnetic induction as,

Whenever a conductor is placed in a varying magnetic field, an electromotive force is induced. If the conductor circuit is closed, a current is induced which is called induced current.

Mentioned here are a few ways to change the magnetic field intensity in a closed loop:

- By rotating the coil relative to the magnet.
- By moving the coil into or out of the magnetic field.
- By changing the area of a coil placed in the magnetic field.
- By moving a magnet towards or away from the coil.



Faraday's Second Law of Electromagnetic Induction

Faraday's second law of electromagnetic induction states that

The induced emf in a coil is equal to the rate of change of flux linkage.

The flux is the product of the number of turns in the coil and the flux associated with the coil.

The formula of Faraday's law is given below:

$$\varepsilon = -N \frac{d\phi}{dt}$$

Where,



- ε is the electromotive force
- Φ is the magnetic flux
- N is the number of turns

The negative sign indicates that the direction of the induced emf and change in the direction of magnetic fields have opposite signs.

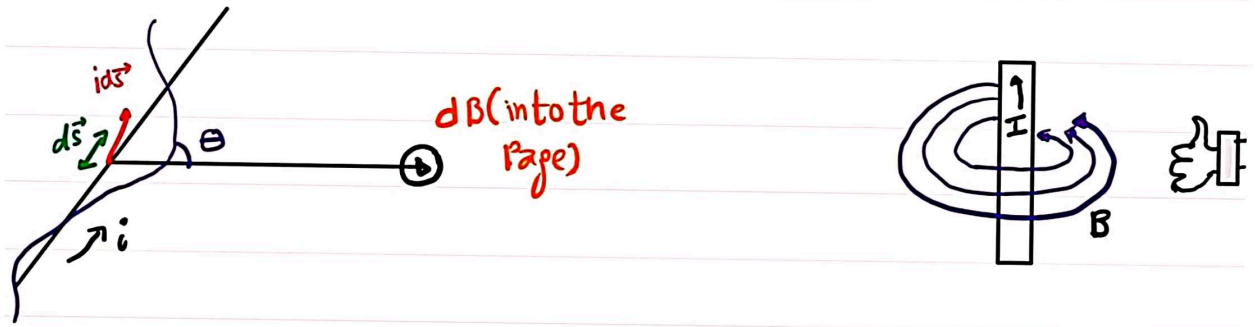
Additionally, there is another key law known as Lenz's law that describes electromagnetic induction as well.

It is evident from Maxwell equations that the third equation represents Faraday's Law of Electromagnetic Induction, which is given as

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Biot Savart Law:

Biot-Savart's law is an equation that gives the magnetic field produced due to a current carrying segment. This segment is taken as a vector quantity known as the current element.



Consider a current carrying wire 'i' in a specific direction as shown in the above figure. Take a small element of the wire of length ds . The direction of this element is along that of the current so that it forms a vector $i ds$.

To know the magnetic field produced at a point due to this small element, one can apply Biot-Savart's Law. Let the position vector of the point in question drawn from the current element be r and the angle between the two be θ . Then,

$$|dB| = \left(\frac{\mu_0}{4\pi}\right) \left(\frac{Idl \sin\theta}{r^2}\right)$$

Where

- μ_0 is the permeability of free space and is equal to $4\pi \times 10^{-7} \text{ TmA}^{-1}$.

The direction of the magnetic field is always in a plane perpendicular to the line of element and position vector. It is given by the right-hand thumb rule where the thumb points to the direction of conventional current and the other fingers show the magnetic field's direction.

In the figure shown above, the direction of the magnetic field is pointing into the page.

Applications of Biot-Savart's Law

Some of Biot-Savart's Law applications are given below.

- We can use Biot-Savart law to calculate magnetic responses even at the atomic or molecular level.
- It is also used in aerodynamic theory to calculate the velocity induced by vortex lines.

Importance of Biot-Savart Law

Following are the importance of Biot-Savart law:

- Biot-Savart law is similar to the Coulomb's law in electrostatics.
- The law is applicable for very small conductors too which carry current.
- The law is applicable for symmetrical current distribution.

Poynting Theorem:

$$\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2}$$

$$\boxed{\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2}}$$

Comparing these
equation.

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

$$\psi = \vec{H}, \quad \frac{1}{v^2} = \mu_0 \epsilon_0.$$

→ Poynting's Theorem:

→ In a uniform plane wave at any point product of Electric Field $\vec{E}$ and magnetic Field Intensity ($\vec{H}$) is a measure of energy flow at that point.

$$\boxed{\vec{P} = \vec{E} \cdot \vec{H}}$$

J.H. Poynting in 1884 gave the following expression.

$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s} = - \underbrace{\int_V (\vec{E} \cdot \vec{J}) dv}_{\text{Power loss}} - \underbrace{\frac{\partial}{\partial t} \int_V \left(\frac{\epsilon E^2}{2} + \frac{\mu H^2}{2} \right) dv}_{\text{Rate of decrease of Stored Energy.}}$$

energy flow per unit area
Power loss
Rate of decrease of Stored Energy.

Now from eq(3) and eq(4) of Maxwell we have,

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad - (1)$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad - (2)$$

Now taking $(\cdot)$ product of equation (2) with $\vec{E}$ we have,

$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \quad - (3)$$

Now from vector Identity we have

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H})$$

$$\vec{E} \cdot (\vec{\nabla} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H})$$

Put this value in eq ③ we have.

$$\vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}$$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \left(\vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \right)$$

Use 2nd Equation of Maxwell we get.

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\vec{H} \cdot \left(\frac{\partial \vec{B}}{\partial t} \right) - \left(\vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \right)$$

$$\vec{B} = \mu_0 \vec{H} \quad , \quad \vec{D} = \epsilon_0 \vec{E}$$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\mu_0 \vec{H} \cdot \left(\frac{\partial \vec{H}}{\partial t} \right) - \left(\vec{E} \cdot \vec{J} + \epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \right)$$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\vec{E} \cdot \vec{J} - \left[\mu_0 \vec{H} \cdot \left(\frac{\partial \vec{H}}{\partial t} \right) + \epsilon_0 \vec{E} \cdot \left(\frac{\partial \vec{E}}{\partial t} \right) \right] \quad (4)$$

$$(1) \quad \frac{\partial}{\partial t} (E^2) = \frac{\partial}{\partial t} (\overset{(1)}{\vec{E}} \cdot \overset{(2)}{\vec{E}}) = \underbrace{\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} + \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}}_{\text{}} \quad \checkmark \quad \checkmark \quad \checkmark$$

$$\frac{\partial}{\partial t} (E^2) = 2 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$\therefore \boxed{\vec{E} \cdot \frac{\partial \vec{E}}{\partial t} = \frac{1}{2} \frac{\partial (E^2)}{\partial t}} \quad (5)$$

$$\textcircled{2} \quad \frac{\partial}{\partial t} (H^2) = \frac{\partial}{\partial t} (\vec{H} \cdot \vec{H}) = \vec{H} \cdot \frac{\partial \vec{H}}{\partial t} + \vec{H} \cdot \frac{\partial \vec{H}}{\partial t}$$

$$\frac{\partial}{\partial t} (H^2) = 2 \vec{H} \cdot \frac{\partial \vec{H}}{\partial t}$$

$$\boxed{\vec{H} \cdot \frac{\partial \vec{H}}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} (H^2)} \quad - \textcircled{6}$$

Substitute eq $\textcircled{5}$ and eq $\textcircled{6}$ in eq $\textcircled{4}$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\vec{E} \cdot \vec{J} - \frac{1}{2} \mu_0 \frac{\partial}{\partial t} (H^2) - \frac{1}{2} \epsilon_0 \frac{\partial}{\partial t} (E^2) \quad - \textcircled{7}$$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\vec{E} \cdot \vec{J} - \frac{\partial}{\partial t} \left[\frac{\mu_0 H^2}{2} + \frac{\epsilon_0 E^2}{2} \right] \quad - \textcircled{8}$$

Apply volume integral in both sides.

$$\int_V \vec{\nabla} \cdot (\vec{E} \times \vec{H}) \cdot d\vec{v} = - \int_V (\vec{E} \cdot \vec{J}) \cdot d\vec{v} - \frac{\partial}{\partial t} \int_V \left(\frac{\mu_0 H^2}{2} + \frac{\epsilon_0 E^2}{2} \right) d\vec{v}.$$

Apply Gauss divergence theorem on left side of eq we get.

$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{S} = - \int_V (\vec{E} \cdot \vec{J}) \cdot d\vec{v} - \frac{\partial}{\partial t} \int_V \left(\frac{\mu_0 H^2}{2} + \frac{\epsilon_0 E^2}{2} \right) d\vec{v}.$$

Hence proved.

→ Gauss Divergence Theorem:

According to this theorem,

$$\int_V (\nabla \cdot \vec{F}) \cdot dV = \oint_S \vec{F} \cdot d\vec{s}$$

$dV \rightarrow$ volume element, $d\vec{s} \rightarrow$ surface element.

This theorem states that integral of a divergence over a volume is equal to value of function at the surface.

→ Since the boundary of volume is always a closed surface, RHS is integral over the closed surface.

→ This theorem converts volume integral into the surface integral. Therefore this theorem is very useful in the situations where it is difficult to calculate volume integral.

(b) Stoke's Theorem:

$$\oint_S (\nabla \times \vec{F}) \cdot d\vec{s} = \oint_C \vec{F} \cdot d\vec{r}$$

This theorem states that the integral of curl of a vector function over a closed surface S is equal to line integral of the function over the element $d\vec{r}$.

→ This theorem converts surface integral to line integral. LHS $\rightarrow$ surface integral
RHS $\rightarrow$ line integral.

→ This theorem converts surface integral to line integral. LHS → surface integral
RHS → line integral.

→ The following theorem is useful in situation when it is difficult to calculate surface integral.

→ Maxwell's Equation in Free Spaces.

we know that Maxwell Equations is given by

$$\textcircled{1} \quad \vec{\nabla} \cdot \vec{D} = \rho$$

$$\textcircled{2} \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\textcircled{3} \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\textcircled{4} \quad \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

→ Particular case of Electromagnetic phenomenon in free space (or) more generally in perfect dielectric containing no-charges and no conduction current.

→ So from this we can say that the volume charge density ρ and conduction current will become zero; hence the equations of Maxwell in modified form are:

$$\vec{\nabla} \cdot \vec{D} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

So the above equations are Maxwell's Equations in Free space.